CERTIFICATE OF INSTALLATION

Note: This table completed by ~~HERS~~ECC Registry.

Project Name:	Enforcement Agency:
Dwelling Address:	Permit Number:
City and Zip Code:	Permit Application Date:

A. Design Dwelling Unit Water Heating Systems Information (other than HPWH)

This table reports features of the water heating system(s) other than HPWH systems specified on the registered CF1R compliance document for this project.

01	02	03	04	05	06	07	08	09	10
Dwelling Unit Name	Water Heating System ID or Name	Water Heating System Type	Water Heater Type	# of Like (or Identical) Water Heaters in System	Fuel Type	Rated Input Type	Rated Input Value	Dwelling Unit DHW System Distribution Type	Compact Distrib.

A2. Design Dwelling Unit HPWH System Information

This table reports the water heating system(s) that were specified on the registered CF1R compliance document for this project.

01	02	03	04	05	06	06a	07	08	09
Dwelling Unit Name	Water Heating System ID or Name	Modeled Equipment Make and Model	# of Like (or Identical) Water Heaters in System	Tank Location	Exterior Tank Insulation R-value	Tank Volume	Dwelling Unit DHW System Distribution Type	Compact Distribution	Simulated Equipment Make and Model

B. Installed Dwelling Unit Water Heating Systems Information

This table reports features of the water heating system other than HPWH systems installed in this project.

01	02	03	04	05	06	07	08	09	10
Dwelling Unit Name	Water Heating System ID or Name	Water Heating System Type	Water Heater Type	# of Like (or Identical) Water Heaters in System	Fuel Type	Rated Input Type	Rated Input Value	Dwelling Unit DHW System Distribution Type	Compact Distrib.

B2. Installed Dwelling Unit HPWH System Information

This table reports the water heating system(s) installed in this project.

01	02	03	04	05	06	06a	07	08
Dwelling Unit Name	Water Heating System ID or Name	Modeled Equipment Make and Model	# of Like (or Identical) Water Heaters in System	Tank Location	Exterior Tank Insulation R-value	Tank Volume	Dwelling Unit DHW System Distribution Type	Compact Distribution

C. Design Dwelling Unit Water Heating Efficiency Information

This table reports the water heater(s) efficiency features specified on the registered CF1R compliance document for this project.

01	02	03	04	05	06	07
Water Heating System ID or Name	Heating Efficiency Type	Heating Efficiency Value	Standby Loss (%)	Exterior Insulation R-Value	Water Heater Storage Volume (gal)	Tank Location

D. Installed Dwelling Unit Water Heating Efficiency Information

This table reports the water heater(s) efficiency features installed in this project.

01	02	03	04	05	06	07
Water Heating System ID or Name	Heating Efficiency Type	Heating Efficiency Value	Standby Loss (%)	Exterior Insulation R-Value	Water Heater Storage Volume (gal)	Tank Location

E. Installed Water Heater Manufacturer Information

01	02	03
Water Heating System ID or Name	Manufacturer	Model Number

F. Mandatory Measures for ~~all Domestic Hot Water Distribution~~Single Dwelling Systems

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	Equipment shall meet the applicable requirements of the Appliance Efficiency Regulations (Section 110.3(b)1).
02	Unfired storage tanks are insulated with an external R-3.5 or combination of R-16 internal and external Insulation. (Section 110.3(c)3~~4~~).
03	**Domestic hot water piping insulation requirements** (Section 150(J)): See the exceptions to Section 150(J) • All domestic hot water piping shall be insulated as specified in Section 609.12~~1~~ of the California Plumbing Code. • Insulation buried below grade must be installed in a waterproof and non-crushable casing or sleeve. • Pipe insulation shall fit tightly and all elbows and tees shall be fully insulated. • Piping that penetrates framing members shall not be required to have pipe insulation for the distance of the framing penetration. • Piping that penetrates metal framing shall use grommets, plugs, wrapping or other insulating material to assure that no contact is made with the metal framing. Insulation shall butt securely against all framing members. • ~~Piping installed in interior or exterior walls that is surrounded on all sides by at least 1 inch (2.5 cm) of insulation.~~ • ~~Piping installed in crawlspace with a minimum of 1 inches (2.5 cm) of crawlspace insulation above and below.~~ Piping surrounded with a minimum of 1 inch of wall insulation, 2 inches of crawlspace insulation, or 4 inches of attic insulation shall not be required to have pipe insulation. • ~~Piping installed in attics with a minimum of 4 inches (10 cm) of attic insulation on top.~~ • ~~Pipe insulation shall fit tightly and all elbows and tees shall be fully insulated.~~
04	For Gas or Propane Water Heaters: Ensure either a **or** b are installed (Section 150.0(n)) a) A designated space at least 2.5 feet by 2.5 feet and 7 feet tall within 3 feet from the water heater • A dedicated 125V, 20A electrical receptacle connected to the electric panel with a 120/240V 3 conductor, branch circuit rated at 30 amps minimum, within 3 feet from the water heater and is accessible with no obstructions. • The conductor shall be labeled with the word "Spare" on both ends; and • A reserved single pole circuit breaker space next to the circuit breaker next to the branch circuit labeled "Future~~"~~ 240V use" shall be provided. • A condensate drain no more than 2 inches higher than the base of the~~n~~ water heater, and allows for ~~for~~ natural draining without pump assistance. b) A designated space at least 2.5 feet by 2.5 feet and 7 feet tall more than 3 feet from the water heater • A dedicated 240 volt branch circuit shall be installed within 3 feet from the designated space. The • branch circuit shall be rated at 30 amps minimum. The blank cover shall be identified as "240V ready"; and • The main electrical service panel shall have a reserved space to allow for the installation of a double pole circuit breaker for a future HPWH installation. The reserved space shall be permanently marked as "For Future 240V use"; and • Either a dedicated cold water supply, or the cold water supply shall pass through the designated HPWH location just before reaching the gas or propane water heater; and • The hot water supply pipe coming out of the gas or propane water heater shall be routed first through the designated HPWH location before serving any fixtures; and • The hot and cold water piping at the designated HPWH location shall be exposed and readily accessible for future installation of a HPWH; and • A condensate drain no more than 2 inches higher than the base of the installed water heater, and allows natural draining without pump assistance.
05	For Air-Source Heat Pump Water Heaters (HPWH), the following shall be met (Section 110.3(c)7) a) Backup heat is required when inlet air is unconditioned, unless the compressor cutout cut-off temperature is below the Heating Winter Median of Extreme. Backup heat may be internal or external to the HPWH b) Meet **one** of the ventilation requirements below. Minimum volume and opening size requirements shall be the sum of all HPWHs installed within the same space. Compressor capacity shall be determined using AHRI 540 Table 4 reference conditions for refrigeration with the "High" rating test point: a. Installed using a method provided by the manufacturer to meet or exceed the level of performance provided by the ventilation requirements of Section 110.3(c)7B2 through Section 110.3(c)B4.

	b. For HPWH installation without ducts, the installation space shall have a volume not less than the greater of 100 cubic feet per kBtu per hour of compressor capacity, or the minimum volume provided by the manufacturer for this method; or c. For HPWH installation without ducts, the installation space shall be vented to a communicating space via permanent openings, according to the following requirements: i. Communicating space shall meet the minimum volume of Section 110.3(c)7B12 above, minus the volume of the HPWH installation space; and ii. Permanent openings shall consist of a single layer of fixed flat slat louvers or grilles, with a total minimum Net Free Area (NFA) the larger of 125 square inches plus 25 square inches per kBtu per hour of compressor capacity, or the minimum provided by the manufacturer for this method. The permanent openings shall be fully louvered doors or two openings of equal area, one in the upper half of the enclosure and one in the bottom half of the enclosure. The top of the upper opening must be 12 inches or less from the enclosure top and the bottom of the lower vent must be 12 inches or less from the enclosure bottom; or d. For HPWH installations with ducts, the following requirements shall be met: i. The space joined to the installation space via ducts shall meet the minimum volume of Section 110.3(c)7B2 above, minus the volume of the HPWH installation space; and ii. All duct connections and building penetrations shall be sealed; and iii. Exhaust air ducts and all ducts which cross pressure boundaries shall be insulated to minimum of R-6; and iv. Where only the HPWH inlet or outlet is ducted, installation space shall include permanent openings which consist of a single layer of fixed flat slat louvers or grilles in the bottom half of the room, and/or a door undercut. With a ducted inlet, the minimum NFA shall be equal to the cross-sectional area of the duct. With a ducted exhaust, the minimum NFA shall be the larger of 20 square inches or the minimum NFA provided by the manufacturer for this method; and v. Where the inlet and outlet ducts both terminate within the same pressure boundary, airflow from the termination points shall be diverted away from each other

G. Compact Hot Water Distribution ~~(CHWDS)~~ (RA4.4.6)

For dwelling units with multiple systems, enter the master bath distance and kitchen distance to the closest water heater, and enter the average of the furthest fixture to each water heater.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	02	03	04	05	06	07	08	09
Dwelling Name	Number of Stories	Master Bath distance of furthest fixture to Water Heater in feet	Kitchen distance from furthest fixture to Water Heater in feet	Furthest Third furthest fixture to Water Heater in feet (Avg for multiple water heaters)	Weighted Distance	Qualification Distance	Design Compactness Factor	Calculated Compactness Factor

H. Central Parallel Piping Requirements ~~(PP)~~ (RA4.4.4)

Systems that utilize this distribution type shall comply with these requirements.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	Each central manifold has 15 feet or less of pipe between manifold and water heater.
02	For manifolds that include valves, the manifold must be readily accessible in accordance with the plumbing code.
03	Hot water distribution system piping from the manifold to the fixtures and appliances must take the most direct path. For instance, piping from a second story manifold cannot supply the first floor.
04	The hot water distribution piping must be separated by at least 2 inches from any other hot water supply piping, and at least 6 inches from any cold water supply piping. Alternatively, the hot water supply piping must be insulated to the thicknesses shown in TABLE 120.3-A-1.

I. Point of Use Requirements (POU) (RA4.4.5)

Systems that utilize this distribution type shall comply with these requirements.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	All hot water supply pipe run lengths are equal to or less than the maximum values shown below, based on the pipe diameter. If a combination of piping is used in a single run, then one half the allowed length of each size is the maximum installed length. The maximum allowed length of piping for the longest run terminating in: 3/8 inch - For only one pipe size - max length allowed is 15 feet For combination pipe sizes the max allowed length of 3/8-inch piping is 7.5 feet, of 1/2 inch piping is 5 feet, and 3/4 inch piping is 2.5 feet. 1/2 inch - For only one pipe size – max length allowed is 10 feet For combination pipe sizes the allowed length of 1/2-inch piping is 5 feet, and 3/4 inch piping is 2.5 feet. 3/4 inch - For only one pipe size = 5 feet

J. Mandatory Requirements for all Recirculation Systems (RA4.4.7)

Systems that utilize a recirculation system shall comply with these requirements.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	A check valve located between the recirculation pump and the water heater to prevent unintentional recirculation.
02	Piping must take the most direct path between water heater and fixtures.
03	Insulation is not required on the cold water line when it is used as the return.
04	If more than one loop is installed, each loop shall have its own pump and controls.

K. Recirculation Non-Demand Controls Requirements ~~(R-ND)~~ (RA4.4.8)

Systems that utilize this distribution type shall comply with these requirements.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	The active control shall be either: timer, temperature, or time and temperature. Timers shall be set to less than 24 hours. The temperature sensor shall be connected to the piping and to the controls for the pump.

L. Demand Recirculation; Manual Control ~~(R-DRmc)~~ (RA4.4.9)**/Sensor Control ~~Requirements (RDRsc)~~** (RA4.4.10) **Requirements**

Systems that utilize either of these distribution types shall comply with these requirements.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	The system operates "on-demand", meaning that the pump begins to operate shortly before or immediately after hot water draw begins, and stops when the return water temperature reaches a certain threshold value. For Demand Recirculation Manual Control, the pump shall be turned on using a manual switch system. For Demand Recirculation Sensor Control, the pump shall be turned on using a sensor system.
02	The controls shall be located in the kitchen, bathroom, and any hot water fixture location that is at least 20 feet from the water heater.
03	Manual controls may be activ~~ated~~e by wired or wireless mechanisms. Each control shall have standby power of 1 Watt or less.
04	Sensor controls may be activated by wired or wireless mechanisms, including buttons, motion sensors, door switches and flow switches. Each control shall have standby power of 1 Watt or less.
05	Pump and control placement shall meet one of the following criteria: • When a dedicated return line has been installed the pump, controls and thermo-sensor are installed at the end of the supply portion of the recirculation loop; or • The pump and controls are installed on the dedicated return line near the water heater and the thermo-sensor is installed in an accessible location as close to the end of the supply portion of the recirculation loop as possible; or • When the cold water line is used as the return, the pump, demand controls and thermo-sensor shall be installed in an accessible location at the end of supply portion of the hot water distribution line (typically under a sink).
06	After the pump has been activated, the controls shall allow the pump to operate until the water temperature at the thermo-sensor rises to one of the following values: • Not more than 10°F (5.6°C) above the initial temperature of the water in the pipe; or • Not more than 102°F (38.9°C).
07	Controls shall limit operation to no more than 5 minutes following activation.

DOCUMENTATION AUTHOR'S DECLARATION STATEMENT

1. I certify that this Certificate of Installation documentation is accurate and complete.

Documentation Author Name:	Documentation Author Signature:
Documentation Author Company Name:	Date Signed:
Address:	CEA/~~HERS~~ECC Certification Identification (If applicable):
City/State/Zip:	Phone:

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this Certificate of Installation is true and correct.
2. I am either: a) a responsible person eligible under Division 3 of the Business and Professions Code in the applicable classification to accept responsibility for the system design, construction, or installation of features, materials, components, or manufactured devices for the scope of work identified on this Certificate of Installation, and attest to the declarations in this statement, or b) I am an authorized representative of the responsible person and attest to the declarations in this statement on the responsible person's behalf.
3. The constructed or installed features, materials, components or manufactured devices (the installation) identified on this Certificate of Installation conforms to all applicable codes and

regulations and the installation conforms to the requirements given on the Certificate of Compliance, plans, and specifications approved by the enforcement agency.

4. I understand that a registered copy of this Certificate of Installation shall be posted or made available with the building permit(s) issued for the building, and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to ensure this requirement is accomplished.
5. I understand that a registered copy of this Certificate of Installation is required to be included with the documentation the builder provides to the building owner at occupancy, and I will take the necessary steps to ensure this requirement is accomplished.

Responsible Builder/Installer Name:	Responsible Builder/Installer Signature:	
Company Name: (Installing Subcontractor or General Contractor or Builder/Owner)	Position With Company (Title):	
Address:	CSLB License:	
City/State/Zip:	Phone:	Date Signed:

CF2R-PLB-02-E User Instructions

A. Design Dwelling Unit Water Heating Systems Information

This table reports the water heating system features that were specified on the registered CF1R compliance document for this project. This section is for information/verification purposes only and requires no user input.

A2. Design Dwelling Unit HPWH System Information

This table reports the water heating system features that were specified on the registered CF1R compliance document for this project. This section is for information/verification purposes only and requires no user input.

B. Installed Dwelling Unit Water Heating Systems Information

This table reports the water heating system information that is being installed. Require one line for each installed water heater.

1. Dwelling Unit Name - Reference information from Table A.
2. Water Heating System ID or Name – Reference information from Table A.
3. Water Heating System Type – Reference information from Table A. The different kinds of water heating system type are DHW, or Combined Hydronic.
4. Water Heater Type – Reference information from Table A. The different kinds of water heaters are Large/Commercial Storage, Small/Consumer Storage, Residential-Duty Commercial Storage, Heat Pump, Boiler, Large/Commercial Instantaneous, Small/Consumer Instantaneous, Residential-Duty Commercial Instantaneous or Indirect.
5. # of Like (or Identical) Water Heaters in system – Reference information from Table A.
6. Fuel Type – Reference information from Table A. The different kinds of fuel types are heat pump, electric resistance, natural gas, and propane.
7. Rated Input Type – Reference information from Table A. For natural gas and propane, the input type is Btu/hr. For heat pump and electric resistance the input type is kW.
8. Rated Input Value – User input. Numerical value of the rated input. Must be equal to or less than value indicated on the CF1R.
9. Dwelling Unit DHW System Distribution Type - Reference information from Table A.
10. Compact Distribution - Reference information from Table A.

B2. Installed Dwelling Unit HPWH System Information

This table reports the water heating system information that is being installed. Require one line for each installed water heater. Not applicable for central systems.

1. Dwelling Unit Name – Reference information from Table A2.
2. Water Heating System ID or Name – Reference information from Table A2.
3. Modeled Equipment Make and Model – User input must be equal to the value indicated on Table A2 as default and allow user to override with an equivalent system based on the simulated equipment in

Table A2. A2 as default and allow user to override with an equivalent system based on the simulated equipment in Table A2.04

4. # of Like (or Identical) Water Heaters in System – Reference information from Table A2.
5. Tank Location – User input. Must be equal to value indicated in Table A2.
6. Exterior Tank Insulation R-value – User input. Must be equal to or higher than value indicated in Table A2.

6a. Tank Volume – User input must equal reference information on Table A2.

7. Dwelling Unit DHW System Distribution Type –Reference information from Table A2.
8. Compact Distribution — Reference information from Table A2.

C. Design Dwelling Unit Water Heating Efficiency Information

This table reports the water heating system features that were specified on the registered CF1R compliance document for this project. This section is for information/verification purposes only and requires no user input.

D. Installed Dwelling Unit Water Heating Efficiency Information

This table reports the water heating system efficiency features installed in this project.

1. Water Heating System ID or Name – Reference information from Table C.
2. Heating Efficiency Type – Reference information from Table C. Different efficiency types are: Energy Factor, AFUE, UEF and Thermal Efficiency.
3. Heating Efficiency Value – User input must be equal to or higher efficiency than value indicated on Table C.
4. Standby Loss – User input. Must be equal to or less than value indicated in Table C. Value may be N/A if CF1R value is N/A.
5. Exterior Insulation R-Value – User input. Must be equal to or higher than value indicated in Table C. Value may be N/A if CF1R value is N/A.
6. Water Heater Storage Volume (gal) – User input. Must be equal to the value indicated in Table C. Value may be N/A if water heater type is instantaneous with zero storage.
7. Tank location – User input. Must be equal to value indicated in Table C.

E. Installed Water Heater Manufacturer Information

This table reports the manufacturer information of the installed water heater(s). Require one line for each installed water heater. Not applicable for central systems.

1. Water Heating System ID or Name – Reference information from Table B or B2.
2. Manufacturer – User input. Enter the name of the water heater manufacturer.
3. Model Number – User input. Enter the model number of the water heater.

F. Mandatory Measures for ~~all Domestic Hot Water Distribution~~Single Dwelling Systems

This table lists the requirements for all DHW systems. ~~HERS~~ ECC rater must ensure all the requirements on this table are met.

G. Compact Hot Water Distribution ~~Basic~~

If performance compliance is used, this table lists the values used in the performance calculation and require no user input.

If prescriptive compliance is used, fill out this table.

1. Dwelling Name. Reference information from Table A2.
2. Enter the master bath distance of furthest fixture to water heater in feet. For multiple water heaters, enter the distance to the closest water heater.
3. Enter the kitchen distance from furthest fixture to water heater in feet. For multiple water heaters, enter the distance to the closest water heater.
4. Enter furthest third fixtures from fixture to water heater in feet. For multiple water heaters, enter the average of the furthest distance of each water heater.
5. Weighted Distance - Calculated value – no user input required.
6. Qualification Distance - Calculated value – no user input required.

H. Central Parallel Piping Requirements
This table only applies to systems indicated as **Parallel Piping.** In addition to the mandatory requirements in Table J, the installer must ensure the requirements in this table are met.

I. Point of Use Requirements
This table only applies to systems indicated as **Point of Use**. In addition to the mandatory requirements in Table J, the installer must ensure the requirements in this table are met.

J. Mandatory Requirements for all Recirculation System
The requirements of this table apply to all recirculation systems listed below.

K. Recirculation Non-Demand Controls Requirements
This table only applies to systems indicated as **Recirculation Non-demand Controls.** In addition to the mandatory requirements in Table J and M, the installer must ensure the requirements in this table are met.

L. Demand Recirculation Manual Control/Sensor Control Requirements
This table only applies to systems indicated as **Demand Recirculation Manual Control** or **Demand Recirculation Senor Control.** In addition to the mandatory requirements in Table H and K, the installer must ensure the requirements in this table are met.

A. Design Dwelling Unit Water Heating Systems Information (other than HPWH)

This table reports features of the water heating system other than **HPWH** systems that were specified on the registered CF1R compliance document for this project.

<<if CF1R do not have any non-HPWH systems, then display the "section does not apply" message; else require one row of data for each non-HPWH water heater identified on the CF1R>>

01	02	03	04	05	06	07	08	09	10
Dwelling Unit Name	Water Heating System ID or Name	Water Heating System Type	Water Heater Type	# of Like (or Identical) Water Heaters in System	Fuel Type	Rated Input Type	Rated Input Value	Dwelling Unit DHW System Distribution Type	Compact Distrib.
<<reference value from CF1R; if Single Family, then value = Single Family>>	<<reference values from CF1R >>	<<reference values from CF1R allowed values= DHW, Combined Hydronic, Hydronic> >>>	<<reference values from CF1R Allowed values = Heat Pump, Boiler, Indirect, Consumer Instantaneous, Commercial Instantaneous, Consumer Storage, Commercial Storage, Residential-Duty Commercial Storage, or Residential-Duty Commercial Instantaneous>>	<<reference values from CF1R>>	<<reference values from CF1R. Allowed values are Heat Pump, Electric Resistance, Natural Gas, or Propane >>	<<reference value from CF1R; If parent CF1R = CF1R-ADD, ALT or NCB, then value = NA; Elseif parent CF1R = CF1R-PRF; then allowed values: if Fuel Type (A06) = Natural Gas, Propane then value= Btu/Hr; If Fuel Type = Electric Resistance, then value = kW; If Fuel Type = Heat Pump, then value = NA>>	<<if A04 = Heat Pump, then result = NA; If performance, reference value from CF1R-PRF; Else if prescriptive, value = NA>>	<<reference values from CF1R If performance then allowed values are *Standard Distribution System * Point of Use * Parallel Piping *Recirculation System Non-Demand Control * Demand Recirculation Manual Control * Demand Recirculation Sensor Control; Else if prescriptive , Allowed values are *Standard Distribution System *Demand Recirculation * Demand Recirculation Manual Control; Else if CF1R-ALT, then value = NA>>	<<If parent CF1R = NCB-01, then if NCB-01 Field M09 = CompactHotWaterDistributionSystem ~~HERS~~ or CompactHotWaterDistributionSystem Basic or CompactDistributionDrainHeatRecovery~~HERS~~ then value is BasicType else NotCompact; Else if parent CF1R = ALT-01 or ADD-01, then value = NotCompact; Else if information not available from PRF-01, allow user select. Allowed values are *Basic *Expanded *NotCompact; Else use value from PRF-01; >>

A2. Design Dwelling Unit HPWH System Information

This table reports the water heating system(s) that were specified on the registered CF1R compliance document for this project.

<<If CF1R has HPWH, then require table, else display the "section does not apply" message>>

<< require one row of data for each HPWH water heater identified on the CF1R>>

01	02	03	04	05	06	06a	07	08	09
Dwelling Unit Name	Water Heating System ID or Name	Modeled Equipment Make and Model	# of Like (or Identical) Water Heaters in System	Tank Location	Exterior Tank Insulation R-value	Tank Volume	Dwelling Unit DHW System Distribution Type	Compact Distribution	Simulated Equipment Make and Model
<<reference value from CF1R; if Single Family, then value = Single Family>>	<<reference value from CF1R>>	<<references value from CF1R-PRF. Else if prescriptive, display “NEEA Tier 3 or above”>>	<<references values from CF1R >>	<<Reference value from CF1R>>	<<reference values from CF1R-PRF; else NA>>	<<reference values from CF1R-PRF; else if parent CF1R = NCB-01 or ADD-01, then allow user input; else NA>>	<<reference values from CF1R. If performance Allowed values are *Standard Distribution System * Point of Use * Central Parallel Piping *Recirculation System Non-Demand Control * Demand Recirculation Manual Control * Demand Recirculation Sensor Control; Else if prescriptive, Allowed values are *Standard Distribution System *Demand Recirculation * Demand Recirculation Manual Control; Else if CF1R-ALT, then value = NA >>	<<If parent CF1R = NCB-01, then if NCB-01 Field M09 = CompactHotWaterDistributionSystem ~~HERS~~ or CompactHotWaterDistributionSystem Basic or CompactDistributionDrainHeatRecovery~~HERS~~ then value is BasicType else NotCompact ; Else if parent CF1R = ALT-01 or ADD-01, then value = NotCompact ; Else if information not available from PRF-01, allow user select. Allowed values are *Basic *Expanded *NotCompact; Else use value from PRF-01; reference values from CF1R. Allowed values are *Basic *None; if CF1R-ALT, then value = NA >>	<<if performance , hide column from user, needed for equivalency lookup; Reference value from XML; Elseif prescriptive, do not require field>>

B. Installed Dwelling Unit Water Heating Systems Information

This table reports features of the water heating system other than **HPWH** systems installed in this project.

<<if Table A does not apply, then display the "section does not apply" message; else require one row of data for each non-HPWH water heater identified in Table A>>

01	02	03	04	05	06	07	08	09	10
Dwelling Unit Name	Water Heating System ID or Name	Water Heating System Type	Water Heater Type	# of Like (or Identical) Water Heaters in System	Fuel Type	Rated Input Type	Rated Input Value	Dwelling Unit DHW System Distribution Type	Compact Distrib.
<<Reference value from A01 >>	<<Reference value from A02 >>	<<Reference value from A03 >>	<<Reference value from A04>>	<<Reference value from A05>	<< Reference value from A06>	<<Reference value from A07 >>	<<User input value which must pass the following range tests: Value may be NA if A08 is NA. If A06 = Natural Gas or Propane, then If B04 = Commercial Storage, then value must be > 75,000 Btu/hr; If B04 = Consumer Storage, then value must be ≤ 75,000 Btu/hr; If B04 = Commercial Instant, then value must be > 200,000 Btu/hr; If B04 = Consumer Instant, then value must be ≤ 200,000 Btu/hr; Else if B04 = Residential-Duty Commercial Storage, then value must be ≤ 105,000 Btu/hr; Else if A06 = Electric Resistance, then	<<Referenced from A09>>	<< Referenced from A10>>

							If B04 = Commercial Storage or Commercial Instant, then value must be > 12 kW; If B04 = Consumer Storage or Consumer Instant, then value must be ≤ 12 kW; Else if B04 = Residential-Duty Commercial Instantaneous, then value must be ≤ 58.6 kW; If the value passes range test and if A06 = Electric Resistance, it is stored in WaterHeaterElectricFiredRatedInput, Otherwise the value is stored in WaterHeaterGasFiredRatedInput Elseif B04 = Boiler or Indirect, no limit on input value >>		

B2. Installed Dwelling Unit HPWH System Information

This table reports the water heating system(s) installed in this project.

<<If Table A2 does apply, then require table, else display the "section does not apply" message>>

<<require one row of data for each HPWH water heater identified in Table A2>>

01	02	03	04	05	06	06a	07	08
Dwelling Unit Name	Water Heating System ID or Name	Modeled Equipment Make and Model	# of Like (or Identical) Water Heaters in System	Tank Location	Exterior Tank Insulation R-value	Tank volume	Dwelling Unit DHW System Distribution Type	Compact Distribution
<<Reference value from A2-01 >>	<<reference value from A2-02>>	<< If performance, user input is equal to A2-03 as default, and allow user to override with an equivalent system based on the simulated equipment in A2-09; elseif prescriptive newly constructed, If CF1R M03 = 1 and # of bedroom >1, then allow user to enter any 240V HPWH; Else if CF1R M03=1 and # of bedroom <= 1, then allow user to enter any HPWH. Else if M03=2, allow user to enter any Tier 3 model Or higher. >>	<< Reference value from A2-04>>	<<Reference value from A2-05>>	<<User input value; check value must be ≥ A2-06 to comply, else flag non-compliant values and do not allow the doc to be registered. Value may be NA if A2-06 is NA >>	<<User input value; check value must be ≥ A2-06a to comply, else flag non-compliant values and do not allow the doc to be registered. Value may be NA if A2-06a is NA >>	<< Reference values from A2-07>>	<< Reference values from A2-08>>

C. Design Dwelling Unit Water Heating Efficiency Information

This table reports the water heater(s) efficiency features specified on the registered CF1R compliance document for this project.

<<require one row of data for each water heater identified in Section A and A2>>

01	02	03	04	05	06	07
Water Heating System ID or Name	Heating Efficiency Type	Heating Efficiency Value	Standby Loss (%)	Exterior Insulation R-Value	Water Heater Storage Volume (gal)	Tank Location
<<Reference value A02 or A2-02>>	<<reference values from CF1R; if parent CF1R= CF1R-ADD, ALT or NCB, then value = NA; else if parent = CF1R-PRF then allowed values are: *Uniform Energy Factor *AFUE *Thermal Efficiency *Value =NA >>	<< reference values from CF1R-PRF-01; Else = NA>>	<<reference values from CF1R-PRF-01; Else = NA >>	<<reference values from CF1R-PRF-01; Else = NA >>	<<reference value from CF1R as default, but if information is not available then allow user input. NA is allowed . NA is allowed only if Water Heater Type = Consumer Instantaneous, Commercial Instantaneous, or Residential-Duty Commercial Instantaneous >>	<<Reference value from CF1R>>

D. Installed Dwelling Unit Water Heating Efficiency Information

This table reports the water heating system efficiency features installed in this project.

<<require one row of data for each water heater identified in Section C.>>

01	02	03	04	05	06	07
Water Heating System ID or Name	Heating Efficiency Type	Heating Efficiency Value	Standby Loss (%)	Exterior Insulation R-Value	Water Heater Storage Volume (gal)	Tank Location
<<Reference value from C01>>	<< If C02 = N/A then allow user to select from list: *Uniform Energy Factor, *AFUE, or *Thermal Efficiency; else reference value from C02 as default >>	<<User input; check value must be ≥ value in C03 to comply, else flag non-compliant values and do not allow the doc to be registered. Value may be NA if C03 is NA >>	<<User input; check value must be ≤ value in C04 to comply, else flag non-compliant values and do not allow the doc to be registered. Value may be NA if C04 is NA >>	<<User input; check value must be ≥ value in C05 to comply, else flag non-compliant values and do not allow the doc to be registered. Value may be NA if C05 is NA >>	<< User Input must equal reference values from C06. NA is allowed only if Water Heater Type = Consumer Instantaneous or Commercial Instantaneous>>	<<Reference value from C07. Value may be NA if CF1R value is NA >>

E. Installed Water Heater Manufacturer Information <<require one row of data for each water heater identified in Section C.>>		
01	02	03
Water Heating System ID or Name	Manufacturer	Model Number
<<Reference value from B02 or B2-02>>	<<User input>>	<<User input>>

F. Mandatory Measures for ~~all Domestic Hot Water Distribution~~Single Dwelling Systems	
01	Equipment shall meet the applicable requirements of the Appliance Efficiency Regulations (Section 110.3(b)1).
02	Unfired storage tanks are insulated with an external R-3.5 or combination of R-16 internal and external Insulation. (Section 110.3(c)3~~4~~).
03	Pipe Insulation (Section 150(J)) • All domestic hot water piping shall be insulated as specified in Section 609.12~~1~~ of the California Plumbing Code. Insulation buried below grade must be installed in a waterproof and non-crushable casing or sleeve. • Pipe insulation shall fit tightly and all elbows and tees shall be fully insulated. • Piping that penetrates framing members shall not be required to have pipe insulation for the distance of the framing penetration. • Piping that penetrates metal framing shall use grommets, plugs, wrapping or other insulating material to assure that no contact is made with the metal framing. Insulation shall butt securely against all framing members. • ~~Piping installed in interior or exterior walls that is surrounded on all sides by at least 1 inch (2.5 cm) of insulation.~~ • ~~Piping installed in crawlspace with a minimum of 1 inches (2.5 cm) of crawlspace insulation above and below.~~ • Piping surrounded with a minimum of 1 inch of wall insulation, 2 inches of crawlspace insulation, or 4 inches of attic insulation shall not be required to have pipe insulation. • ~~Piping installed in attics with a minimum of 4 inches (10 cm) of attic insulation on top.~~ • ~~Pipe insulation shall fit tightly and all elbows and tees shall be fully insulated.~~
04	For Gas or Propane Water Heaters: Ensure either a **or** b are installed (Section 150.0(n)) a) A designated space at least 2.5 feet by 2.5 feet and 7 feet tall within 3 feet from the water heater • A dedicated 125V, 20A electrical receptacle connected to the electric panel with a 120/240V 3 conductor, ~~10 AWG copper~~ branch circuit rated at 30 amps minimum, within 3 feet from the water heater and is accessible with no obstructions. • The conductor shall be labeled with the word “Spare” on both ends; and • A reserved single pole circuit breaker space next to the circuit breaker next to the branch circuit labeled “Future ”~~-~~240V use” shall be provided. • A condensate drain no more than 2 inches higher than the base of the~~n~~ water heater, and allows ~~for~~ natural draining without pump assistance. b) A designated space at least 2.5 feet by 2.5 feet and 7 feet tall more than 3 feet from the water heater • A dedicated 240 volt branch circuit shall be installed within 3 feet from the designated space. The • branch circuit shall be rated at 30 amps minimum. The blank cover shall be identified as “240V ready”; and • The main electrical service panel shall have a reserved space to allow for the installation of a double pole circuit breaker for a future HPWH installation. The reserved space shall be permanently marked as “For Future 240V use”; and • Either a dedicated cold water supply, or the cold water supply shall pass through the designated HPWH location just before reaching the gas or propane water heater; and • The hot water supply pipe coming out of the gas or propane water heater shall be routed first through the designated HPWH location before serving any fixtures; and • The hot and cold water piping at the designated HPWH location shall be exposed and readily accessible for future installation of a HPWH; and A condensate drain no more than 2 inches higher than the base of the installed water heater, and allows natural draining without pump assistance

05	For Air-Source Heat Pump Water Heaters (HPWH), the following shall be met (Section 110.3(c)7) a) Backup heat is required when inlet air is unconditioned, unless the compressor cutout cut-off temperature is below the Heating Winter Median of Extreme. Backup heat may be internal or external to the HPWH b) Meet **one** of the ventilation requirements below. Minimum volume and opening size requirements shall be the sum of all HPWHs installed within the same space. Compressor capacity shall be determined using AHRI 540 Table 4 reference conditions for refrigeration with the "High" rating test point: a. Installed using a method provided by the manufacturer to meet or exceed the level of performance provided by the ventilation requirements of Section 110.3(c)7B2 through Section 110.3(c)B4. b. For HPWH installation without ducts, the installation space shall have a volume not less than the greater of 100 cubic feet per kBtu per hour of compressor capacity, or the minimum volume provided by the manufacturer for this method; or c. For HPWH installation without ducts, the installation space shall be vented to a communicating space via permanent openings, according to the following requirements: i. Communicating space shall meet the minimum volume of Section 110.3(c)7B12 above, minus the volume of the HPWH installation space; and ii. Permanent openings shall consist of a single layer of fixed flat slat louvers or grilles, with a total minimum Net Free Area (NFA) the larger of 125 square inches plus 25 square inches per kBtu per hour of compressor capacity, or the minimum provided by the manufacturer for this method. The permanent openings shall be fully louvered doors or two openings of equal area, one in the upper half of the enclosure and one in the bottom half of the enclosure. The top of the upper opening must be 12 inches or less from the enclosure top and the bottom of the lower vent must be 12 inches or less from the enclosure bottom; or d. For HPWH installations with ducts, the following requirements shall be met: i. The space joined to the installation space via ducts shall meet the minimum volume of Section 110.3(c)7B2 above, minus the volume of the HPWH installation space; and ii. All duct connections and building penetrations shall be sealed; and iii. Exhaust air ducts and all ducts which cross pressure boundaries shall be insulated to minimum of R-6; and iv. Where only the HPWH inlet or outlet is ducted, installation space shall include permanent openings which consist of a single layer of fixed flat slat louvers or grilles in the bottom half of the room, and/or a door undercut. With a ducted inlet, the minimum NFA shall be equal to the cross-sectional area of the duct. With a ducted exhaust, the minimum NFA shall be the larger of 20 square inches or the minimum NFA provided by the manufacturer for this method; and v. Where the inlet and outlet ducts both terminate within the same pressure boundary, airflow from the termination points shall be diverted away from each other
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.	

G. Compact Hot Water Distribution ~~(CHWDS)~~ (RA4.4.6)

For dwelling units with multiple systems, enter the master bath distance and kitchen distance to the closest water heater, and enter the average of the furthest fixture to each water heater.

<<require one row of data, reporting the longest distances, for each dwelling unit identified in Table B or B2. with B10 or B2-08 = Basic. If no dwelling in B10 or B2-08 = Basic, then display section header and standard “This section does not apply” message>>

01	02	03	04	05	06	07	08	09
Dwelling Name	Number of Stories	Master Bath distance of furthest fixture to Water Heater in feet	Kitchen distance from furthest fixture to Water Heater in feet	Furthest Third furthest fixture to Water Heater in feet (Avg for multiple water heaters)	Weighted Distance	Qualification Distance	Design Compactness Factor	Calculated Compactness Factor
<<Reference value from A01 or A2-01 >>	<< if prescriptive or G03/G04/G05 are not in performance file, user select from list: 1, 2, 3; Else value = N/A >>	<<Reference Value from CF1R-PRF; Else if prescriptive compliance or not available in XML, user input>>	<<Reference Value from CF1R-PRF; Else if prescriptive compliance or not available in XML, user input>>	<<Reference Value from CF1R-PRF; Else if prescriptive compliance or not available in XML, user input>>	<<Reference value from CF1R-PRF; else if prescriptive or not available in XML and A09 or A2-07 = Standard Distribution System, then value = (G03*0.4) +(G04*0.4)+(G05*0.2); else if A09 or A2-07 = Demand Recirculation Manual Control, then value = G05>>	<< Reference Value from CF1R-PRF; Else if prescriptive compliance or not available in XML, value = ((a+b *CFA)/n) >> *Where:* *a, b = Qualification distance coefficients from Table 4.4.6-2 below,* *CFA = Conditioned floor area of the dwelling unit (ft^2) from CF1R, and* *n = Number of water heaters in the dwelling unit from A05 or A2-04 (unitless).*	<<Reference value from CF1R-PRF; elseif prescriptive, then value = n/a>>	<<Calculated value = 0.3 + 0.4 * (G06/G07), if performance , value must be ≤ G08; elseif prescriptive, value must be ≤ 0.7 >>

The responsible person’s signature on this compliance document affirms that all applicable requirements in this table have been met

Table 4.4.6-2: Coefficients for the Qualification Distance Calculation
<< do not show table, only use for equation in G07>>

	Coefficient a		Coefficient b	
Building Type	**Non-Recirculating**	**Recirculating**	**Non-Recirculating**	**Recirculating**
Single Family	Use when distribution type (A09 Or A2-07) is *Standard Distribution System	Use when distribution type (A09 Or A2-07) is * Demand Recirculation Manual Control		
One story	10	22.7	0.0095	0.0099
Two story	15	11.5	0.0045	0.0095
Three story	10	0.5	0.0030	0.014

H. Central Parallel Piping Requirements ~~(PP)~~ (RA4.4.4)
Systems that utilize this distribution type shall comply with these requirements.
<<If A09 or A2-07 = "Parallel Piping", then display this entire table, else display "section does not apply" message >>

01	Each central manifold has 15 feet or less of pipe between manifold and water heater
02	For manifolds that include valves, the manifold must be readily accessible in accordance with the plumbing code. (RA 4.4.4)
03	Hot water distribution system piping from the manifold to the fixtures and appliances must take the most direct path. For instance, piping from a second story manifold cannot supply the first floor.
04	The hot water distribution piping must be separated by at least 2 inches from any other hot water supply piping, and at least 6 inches from any cold water supply piping. Alternatively, the hot water supply piping must be insulated to the thicknesses shown in TABLE 120.3-A-1.
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.	

I. Point of Use Requirements (POU) (RA4.4.5)
Systems that utilize this distribution type shall comply with these requirements.
<<If A09 or A2-07 = "Point of Use", then display this entire table, else display "section does not apply" message>>

01	All hot water supply pipe run lengths are equal to or less than the maximum values shown below, based on the pipe diameter. If a combination of piping is used in a single run, then one half the allowed length of each size is the maximum installed length. The maximum allowed length of piping for the longest run terminating in: 3/8 inch - For only one pipe size - max length allowed is 15 feet For combination pipe sizes the max allowed length of 3/8-inch piping is 7.5 feet, of 1/2 inch piping is 5 feet, and 3/4 inch piping is 2.5 feet. 1/2 inch - For only one pipe size – max length allowed is 10 feet For combination pipe sizes the allowed length of 1/2 inch piping is 5 feet, and 3/4 inch piping is 2.5 feet. 3/4 inch - For only one pipe size = 5 feet
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.	

J. Mandatory Requirements for all Recirculation System (RA4.4.7)
Systems that utilize a recirculation system shall comply with these requirements.
<<If A09 or A2-07 = "Recirculation System Non-Demand Control", "Demand Recirculation Manual Control", or "Demand Recirculation Sensor Control", then display this entire table, else display "section does not apply" message>>

01	A check valve located between the recirculation pump and the water heater to prevent unintentional recirculation.
02	Piping must take most the direct path between water heater and fixtures.
03	Insulation is not required on the cold water line when it is used as the return.
04	If more than one loop is installed each loop shall have its own pump and controls.
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.	

K. Recirculation Non-Demand Controls Requirements ~~(R-ND)~~ (RA4.4.8) Systems that utilize this distribution type shall comply with these requirements. <<If A09 or A2-07 = "Recirculation System Non-Demand Control", then display this entire table, else display "section does not apply" message >>	
01	The active control shall be either: timer, temperature, or time and temperature. Timers shall be set to less than 24 hours. The temperature sensor shall be connected to the piping and to the controls for the pump.
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.	

L. Demand Recirculation; Manual Control ~~(R-DRmc)~~ (RA4.4.9)/Sensor Control Requirements ~~(RDRsc)~~ (RA4.4.10) Systems that utilize either of these distribution types shall comply with these requirements. <<If A09 or A2-07 = "Demand Recirculation Manual Control" or "Demand Recirculation Sensor Control", then display this entire table, else display "section does not apply" message >>	
01	The system operates "on-demand", meaning that the pump begins to operate shortly before or immediately after hot water draw begins, and stops when the return water temperature reaches a certain threshold value. For Demand Recirculation Manual Control, the pump shall be turned on using a manual switch system. For Demand Recirculation Sensor Control, the pump shall be turned on using a sensor system.
02	The controls shall be located in the kitchen, bathroom, and any hot water fixture location that is at least 20 feet from the water heater.
03	Manual controls may be activated by wired or wireless mechanisms. Each control shall have standby power of 1 Watt or less.
04	Sensor controls may be activated by wired or wireless mechanisms, including buttons, motion sensors, door switches and flow switches. Each control shall have standby power of 1 Watt or less.
05	Pump and control placement shall meet one of the following criteria: • When a dedicated return line has been installed the pump, controls and thermo-sensor are installed at the end of the supply portion of the recirculation loop; or • The pump and controls are installed on the dedicated return line near the water heater and the thermo-sensor is installed in an accessible location as close to the end of the supply portion of the recirculation loop as possible; or • When the cold water line is used as the return, the pump, demand controls and thermo-sensor shall be installed in an accessible location at the end of supply portion of the hot water distribution line (typically under a sink).
06	After the pump has been activated, the controls shall allow the pump to operate until the water temperature at the thermo-sensor rises to one of the following values: • Not more than 10°F (5.6°C) above the initial temperature of the water in the pipe; or • Not more than 102°F (38.9°C).
07	Controls shall limit operation to no more than 5 minutes following activation.
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.	